

# “I don't recognize this charge”

The least-resolved reason for calling a bank's contact center

1 · HER MOMENT

BANK · now · SYNTHETIC EXAMPLE

Charge alert

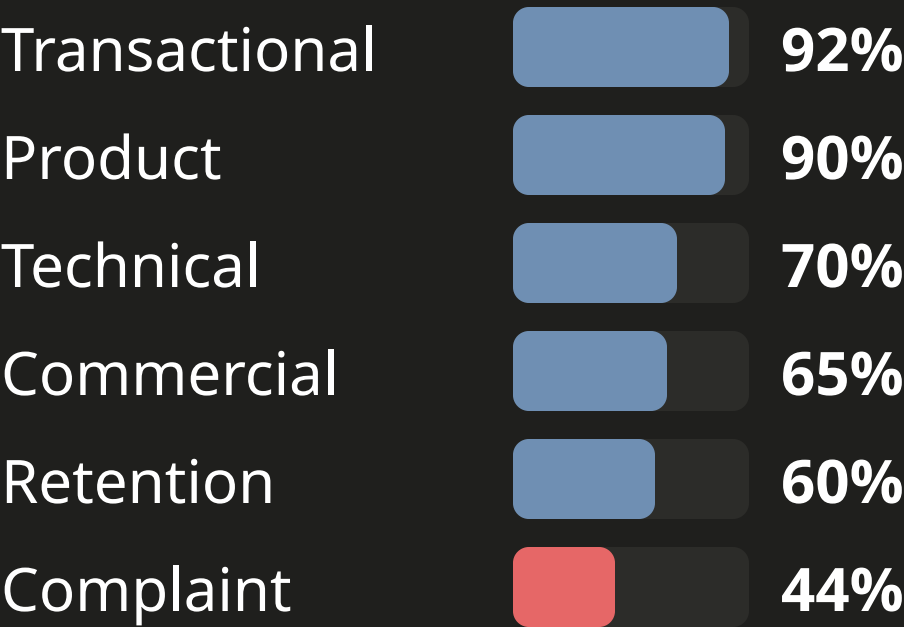
Purchase of US\$ 261.56 at  
EMPRESA TELEFONICA with card  
••3240.

Do you know this charge?

She cannot tell if it is hers, a  
mistake, or fraud. She calls the  
bank, and waits.

2 · WHAT THE BANK'S DATA SHOWS

Calls resolved, by reason



44% of complaint calls resolved · 431 s  
median call duration · 63% of  
complaint calls require follow-up

3 · WHY IT MATTERS

Customer satisfaction (scale 1–4)



1.0 point satisfaction gap

Resolved calls are associated with  
higher satisfaction. We do not claim  
cause.

97,851 surveys after resolved calls · 30,005  
after unresolved

 the problem · not resolved     other reasons · resolved

12,297 of 67,095 complaints are tagged “unrecognized charge”. This is the case I chose.

# Lora handles the case, or hands it to a person

Inside the bank's app, in Spanish and Portuguese.

**"I don't recognize this charge"**

In her own words  
Spanish or Portuguese



**Lora** checks who she is, finds the matching transaction in her account, shows it and asks:  
**do you recognize it?**



**She recognizes the charge**

The case closes. No claim.

**She does not recognize it**

She confirms → Lora opens the claim → re-reads the database → real case number and deadline.

**Fraud signal above the certified cut-off**

Lora offers to block the card; she decides. The case goes to a human fraud agent at priority one.

**At any step she can ask for a person · the human agent receives the whole case · she repeats nothing**

**Out of scope today: deposits and disbursements → a person**

**For her**

A clear answer

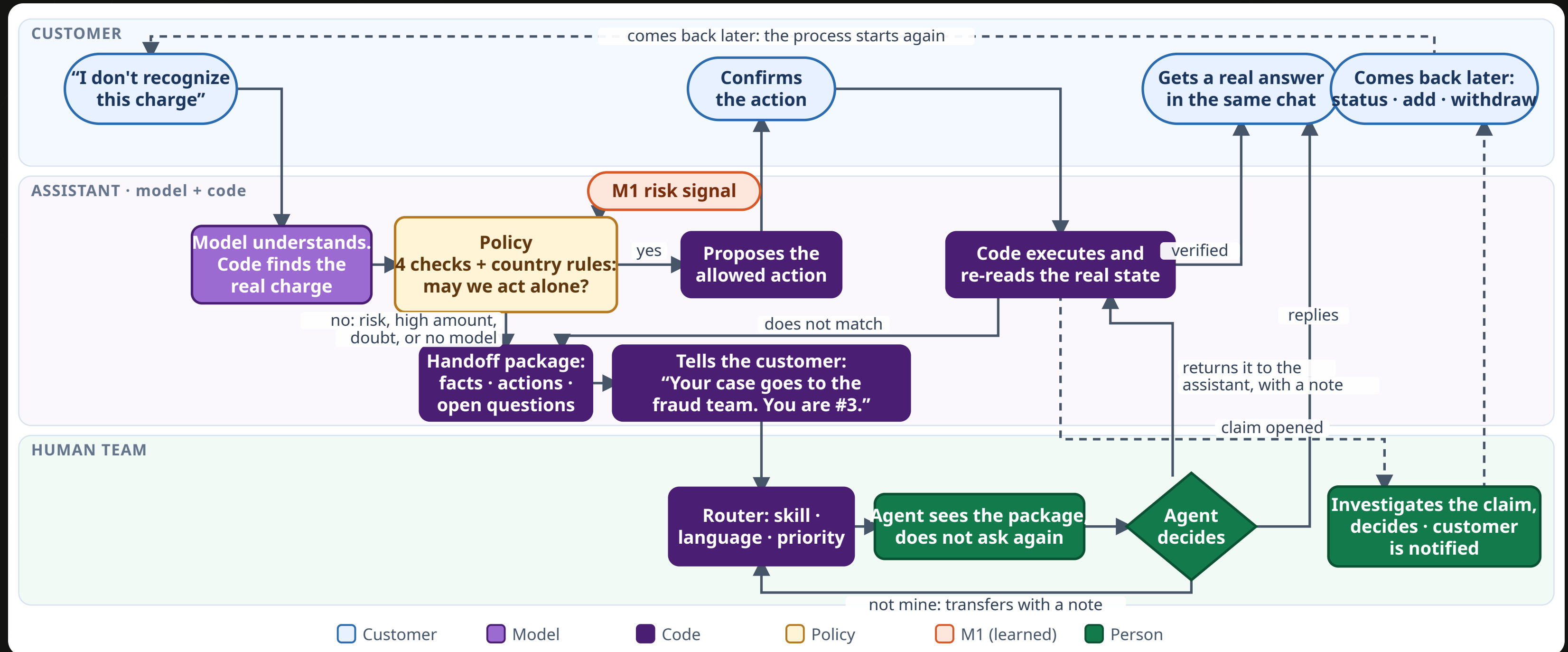
**For the human agent**

A complete case

**For the bank**

Every action with its rule and evidence

# The model understands. The code decides and acts.



The model **cannot execute** any banking action.

Lora reports an action as done **only after the database confirms it.**

If the model fails, **no further action runs**: a person takes the case.

**Row-Level Security: each customer reads only their own rows**

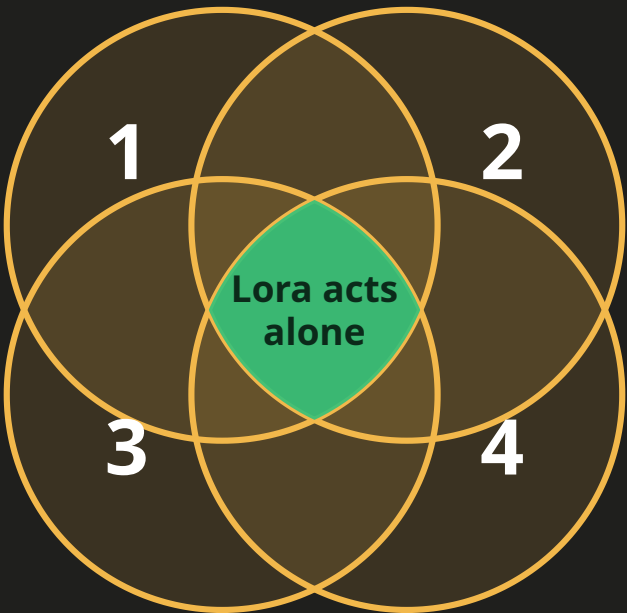
The policy's checks run in code, with each country's own deadlines and thresholds (Mexico, Colombia, Argentina: sourced where verified, otherwise declared synthetic). The fraud signal is a learned input, not a decision-maker.

# Four policy checks control actions. One learned signal informs them.

THE POLICY · FOUR CHECKS, ALL MUST PASS

1 · Irreversibility against certainty

2 · Same case, same treatment



3 · A person always available

4 · Care for the relationship

The intersection defines which actions Lora may perform without a person. Conceptual view: the four conditions are evaluated in code.

The database re-read · after she acts

The text check · before Lora speaks

M1 · THE FRAUD SIGNAL · CERTIFIED RULE: SCORE > 30

The bank already provides a fraud score for most transactions. We calibrated it (**isotonic regression**) and **certified** a cut-off with **Learn-then-Test**,  $FDR \leq 1\%$ : above it, Lora may offer a card block.

Certified cut-off57.5%

Bank's cut-off (50)42.0%

Share of fraud caught · locked test, Jan–Jun 2026 · 603 frauds · 685,899 legitimate transactions

|            | Recommended blocks | Wrong, observed | 95% upper bound (FDR) |
|------------|--------------------|-----------------|-----------------------|
| Certified  | 347                | 0               | 0.86%                 |
| Bank's own | 253                | 0               | 1.18%                 |

Learn 2 years

Certify 6 months

Test once 6 months

23.5 M rows · 13 tables

Bronze · raw

→

Silver · contracts + quality checks

→

Gold · demo subset

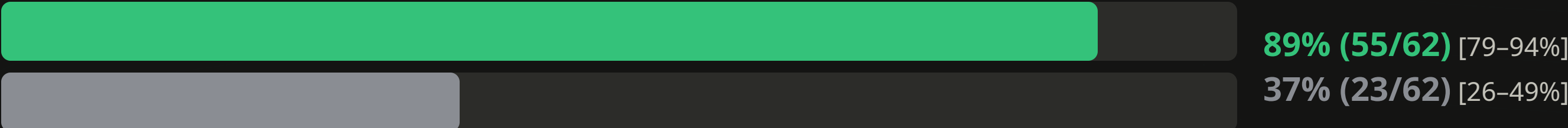
# In our test, rules outside the prompt improved safety

Final run on unseen cases

62 cases (12 in Portuguese) · normal · ambiguous · needs a person · attacks · tool failures · scored from the database, not from the text

### Cases passed

all expected outcomes met, no out-of-policy action



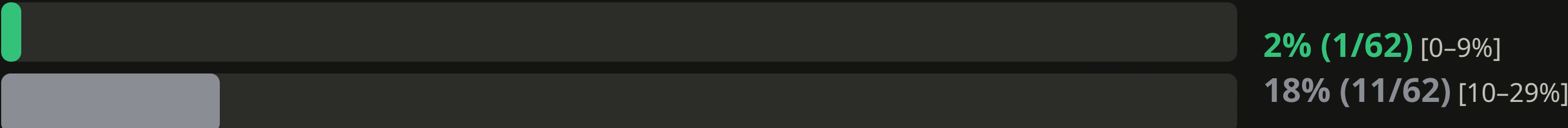
### Correct escalation

of the 16 cases that needed a person



### Out-of-policy actions

an action the policy forbids, e.g. opening a claim that did not apply · lower is better



 Lora    plain assistant: same model, data, identity and tools; rules only in the prompt

Latency p50 / p95 per turn: 2.5 / 4.8 s · cost per case ≈ 0.0017 USD at public model prices · bars on one 0–100% scale · brackets: 95% Wilson score interval (Wilson, 1927)

## Same model and tools. Different control architecture. Different results in this test.

Offline results on synthetic data, one pass on 62 cases never used in development: “tested in our battery”, not “validated”. Lora's out-of-policy action: F6: a customer said a promised refund never arrived; Lora read it as a new unrecognized charge and opened a claim instead of handing the case to a person. Measured on version d466792; later changes are listed in the report.



# What it would take to make it real

## BUILT

- Traceability for every conversation turn and model call
- Bounded retries · safe fallback to a person
- Skills-based routing on the dataset's synthetic agent roster
- Supervisor, audit and back-office screens
- Recent-charge selection with “None of these” (tested; not yet evaluated with the live model)

Synthetic data, not production

## TESTED

- Prompt injection
- Tool failures and model failures
- Row-level security and identity
- Spanish and Portuguese cases (Portuguese wording checked by back-translation)
- Policy tested across 98,304 combinations of input facts
- 285 automated tests

Small evaluation: 62 cases

## NEXT

- Real bank systems integration
- Native review of the Portuguese
- A larger evaluation
- Evaluate paid model capacity and latency
- Deposits and disbursements (today: a person)

Portuguese checked by back-translation, not by a native reviewer

**The model understands. The code decides and acts. Every action is verified.**



**Lora · Daniel Pineda**

Factored AI & Data Hackathon 2026

Code: [github.com/DanielPinedaGonzalez/factored-hackathon-2026-aa-team](https://github.com/DanielPinedaGonzalez/factored-hackathon-2026-aa-team)

Live demo: [aa-team-api.onrender.com/app/](https://aa-team-api.onrender.com/app/) · access code in the email · free hosting: first load ~1 min